

Ethanol Blending in Gasoline - Nicaragua

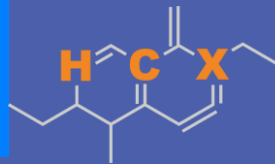
July 2026



**U.S. GRAINS &
BIOPRODUCTS
COUNCIL**

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90



www.faro90.com
+52 55 6122 2255

Ethanol Blending in Latin America

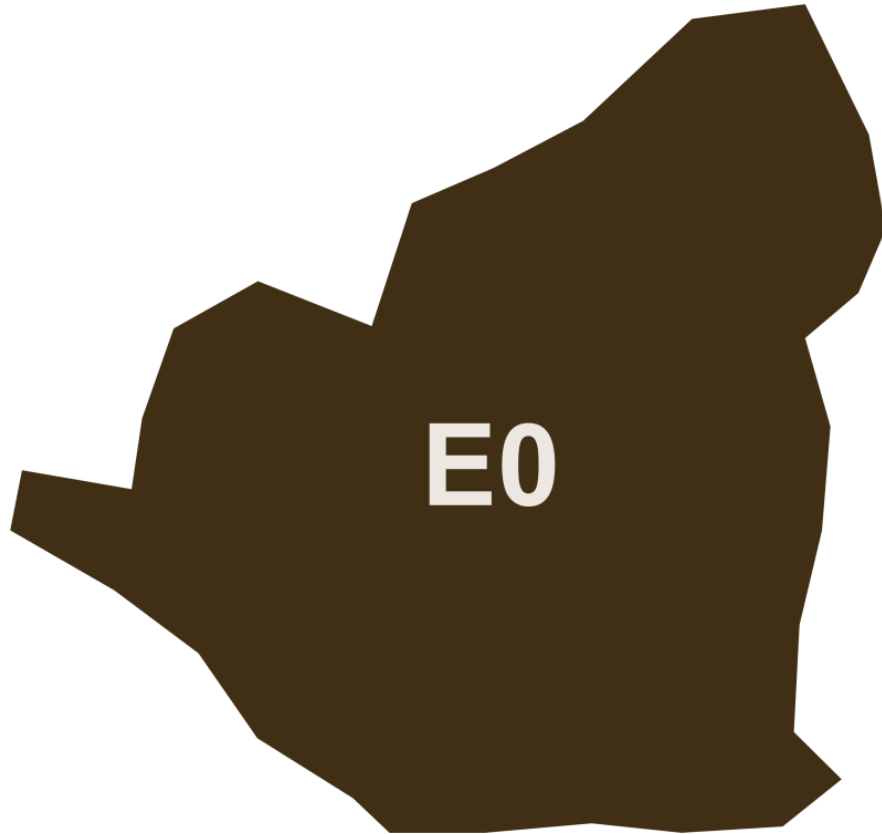
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



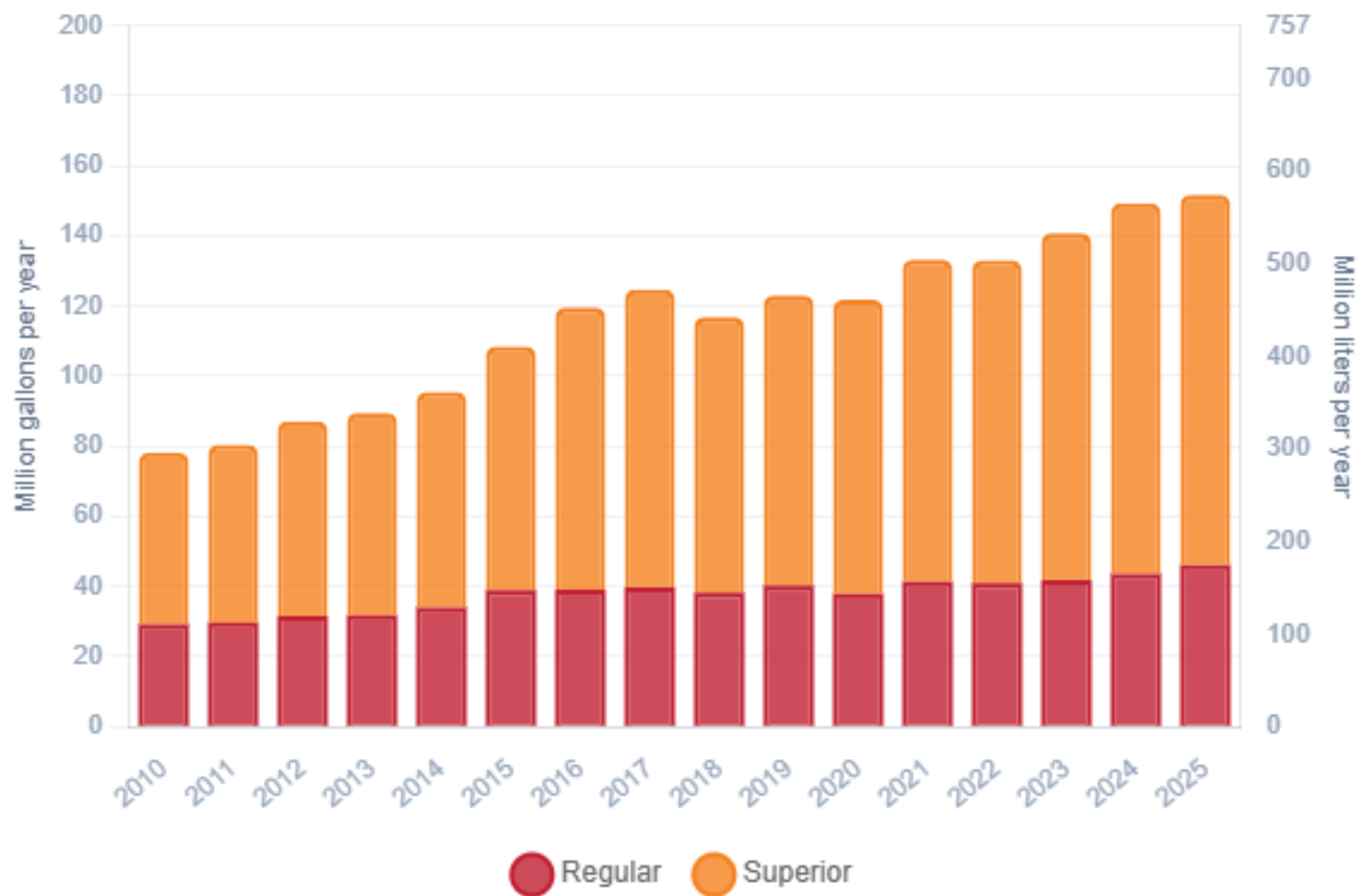
Ethanol Blending in Gasoline - Nicaragua



In 2025, gasoline demand was 151 million gallons (574 million liters). Nicaragua is the only country in Central America that has active refinery capacity, which produced 38% of gasoline national demand. The rest of demand is imported, previously from Venezuela and since 2016 mainly from United States. There are two gasoline grades in the country, regular with 30% and premium with 70% of market share.

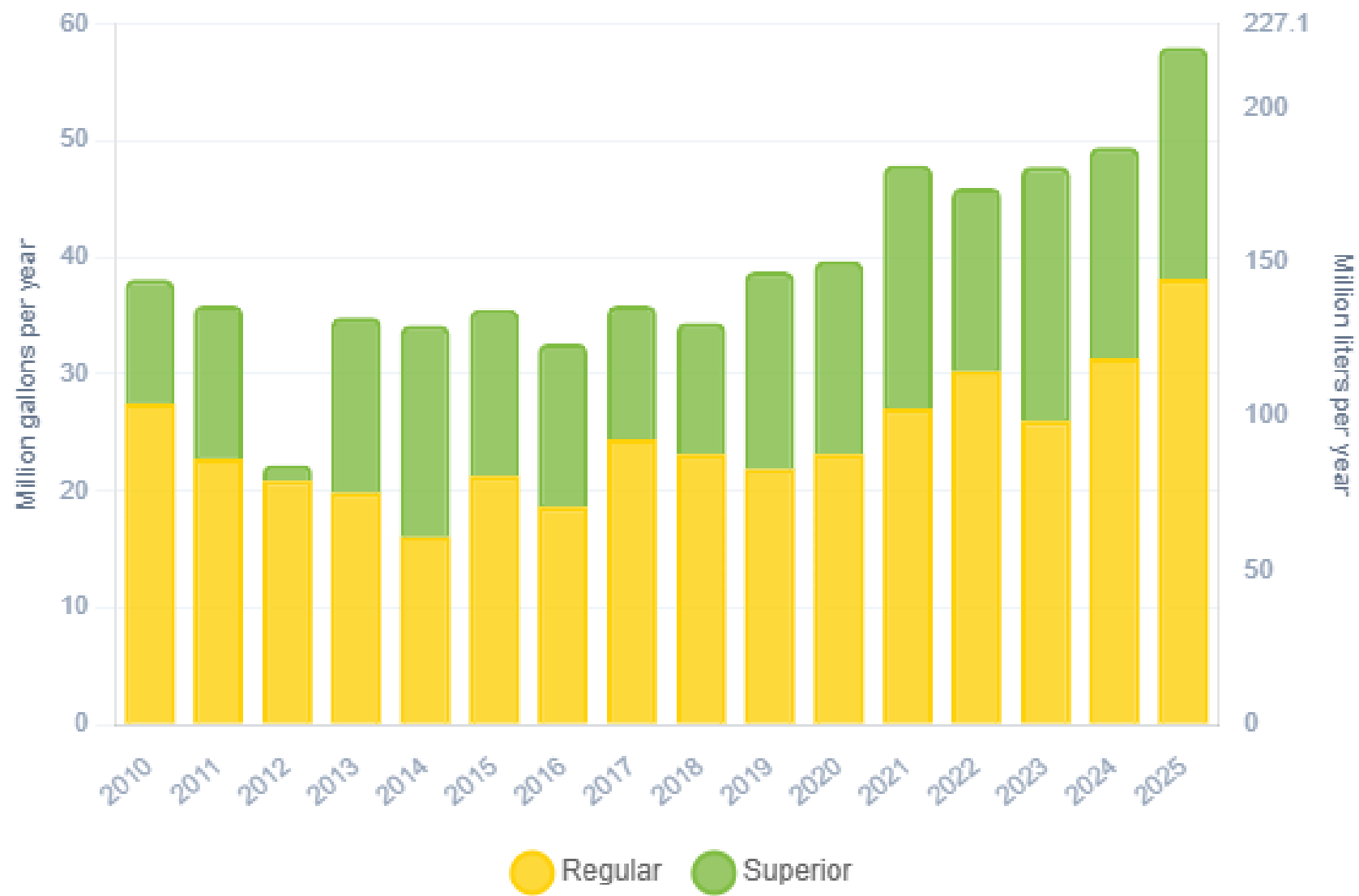
According to the Ministry of Energy and RTCA, Nicaragua has no ethanol mandate.

Gasoline Demand in Nicaragua



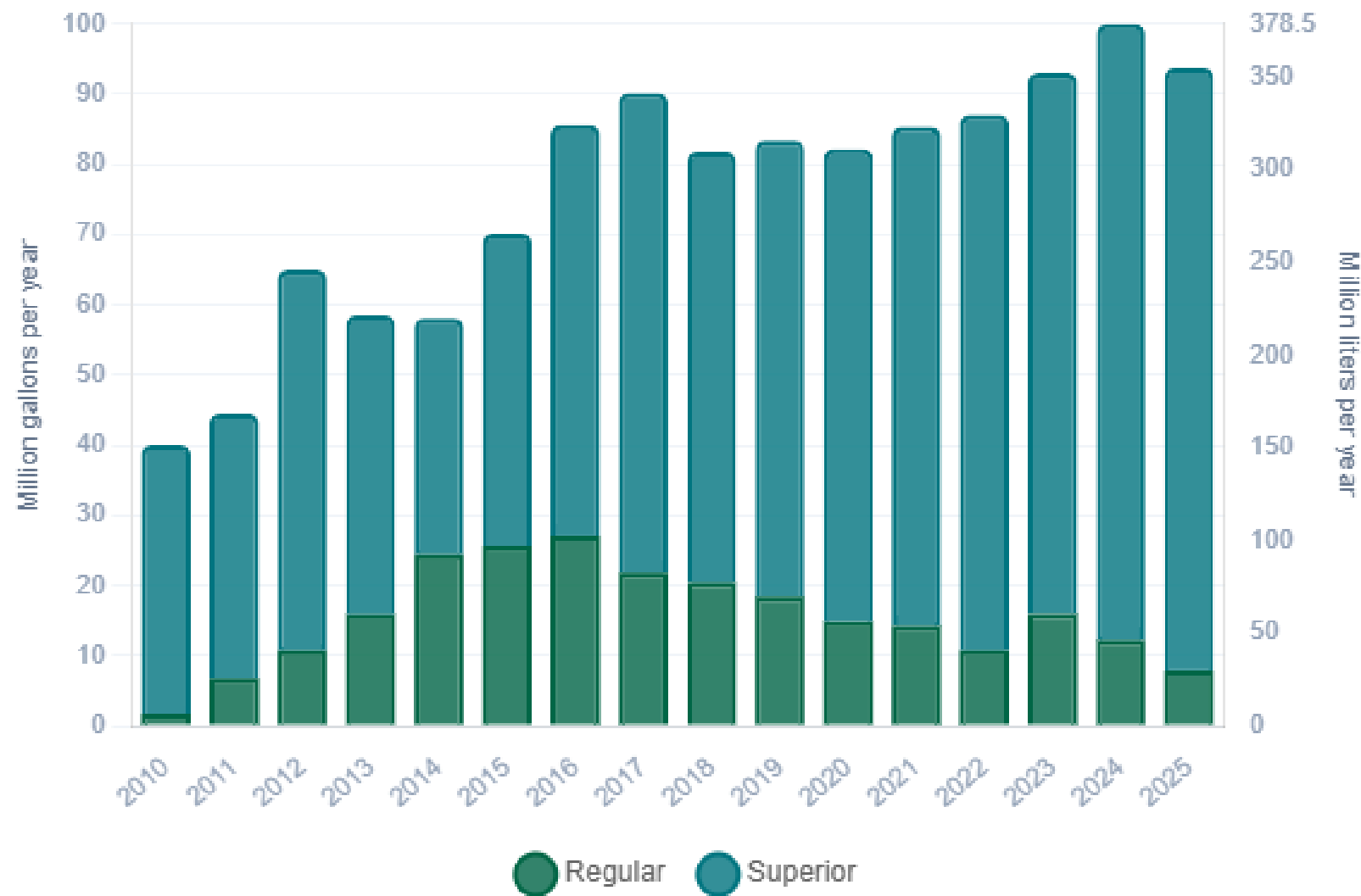
Source: MEM Nicaragua

Gasoline Production in Nicaragua



Source: MEM Nicaragua

Gasoline Imports in Nicaragua



Source: MEM Nicaragua

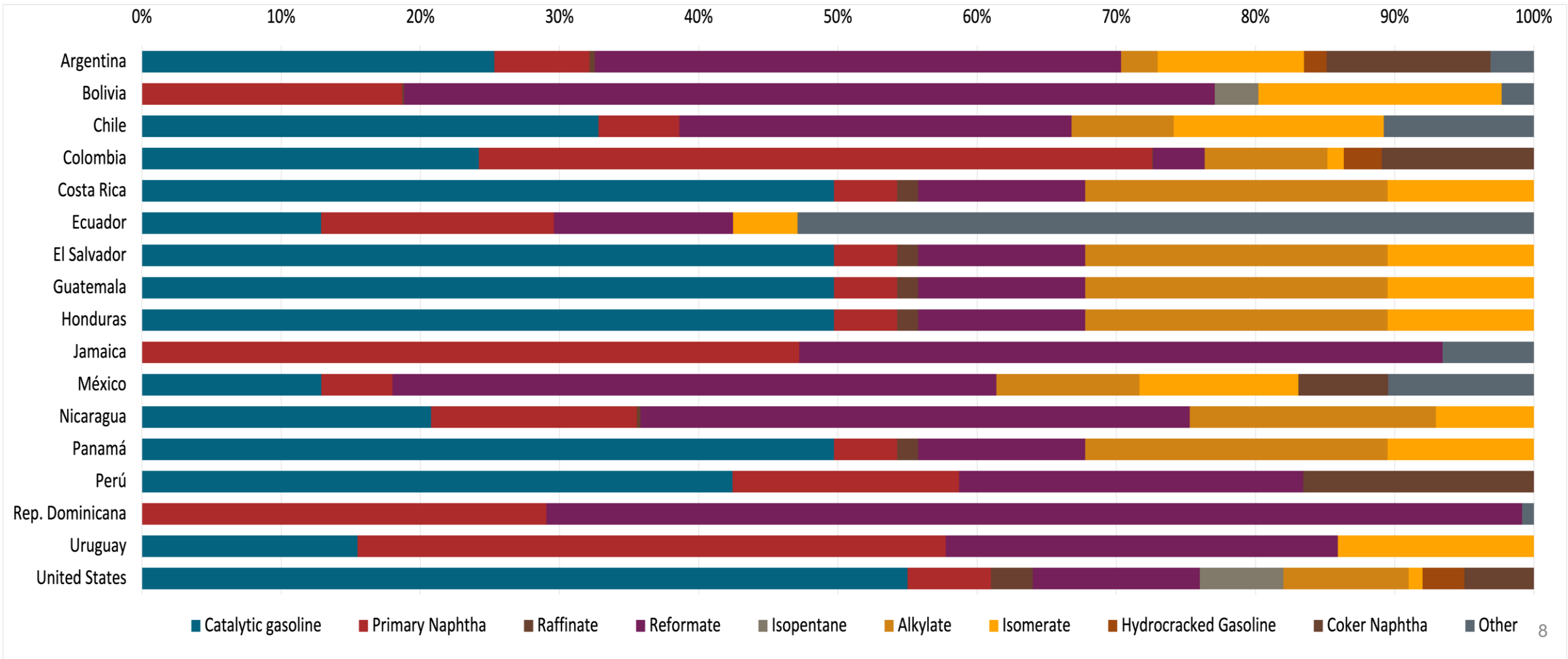
Gasoline Quality in Nicaragua

Name	RTCA 75.01.20:19	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: RTCA

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



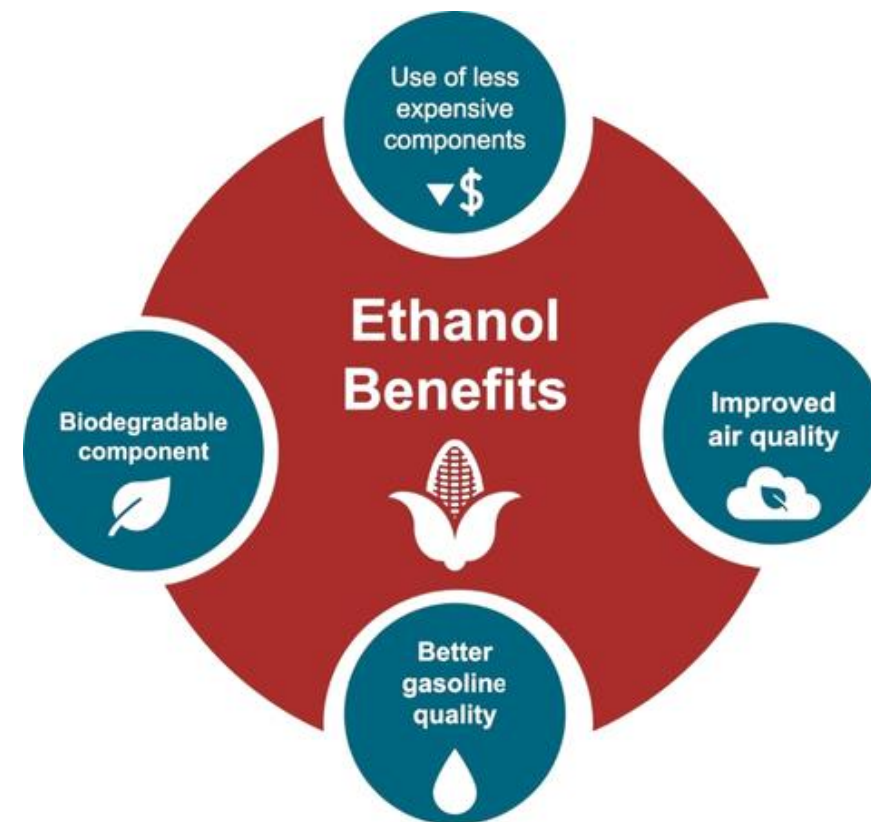
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

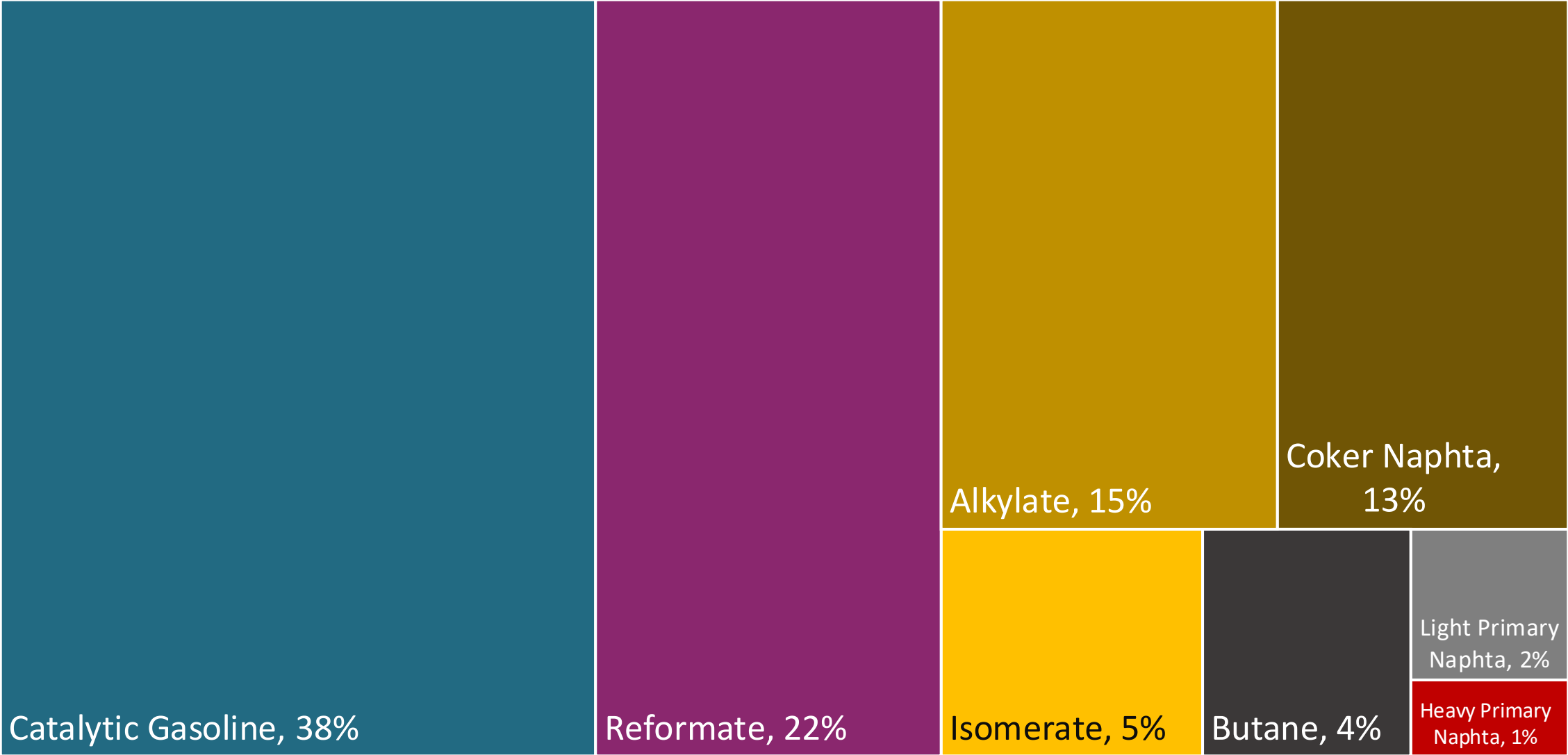
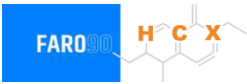
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

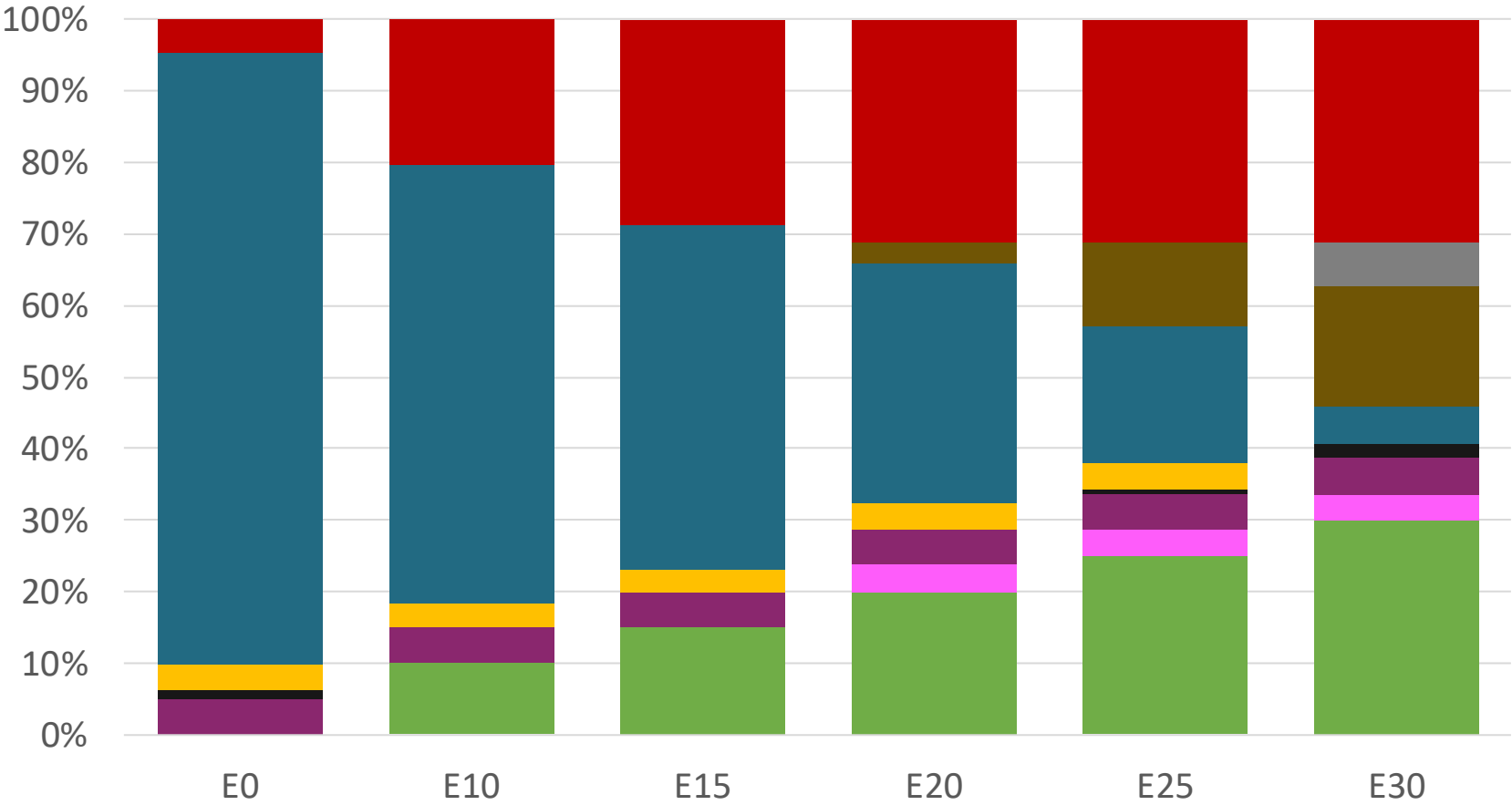
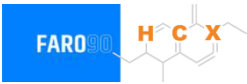


Blending Components - Nicaragua



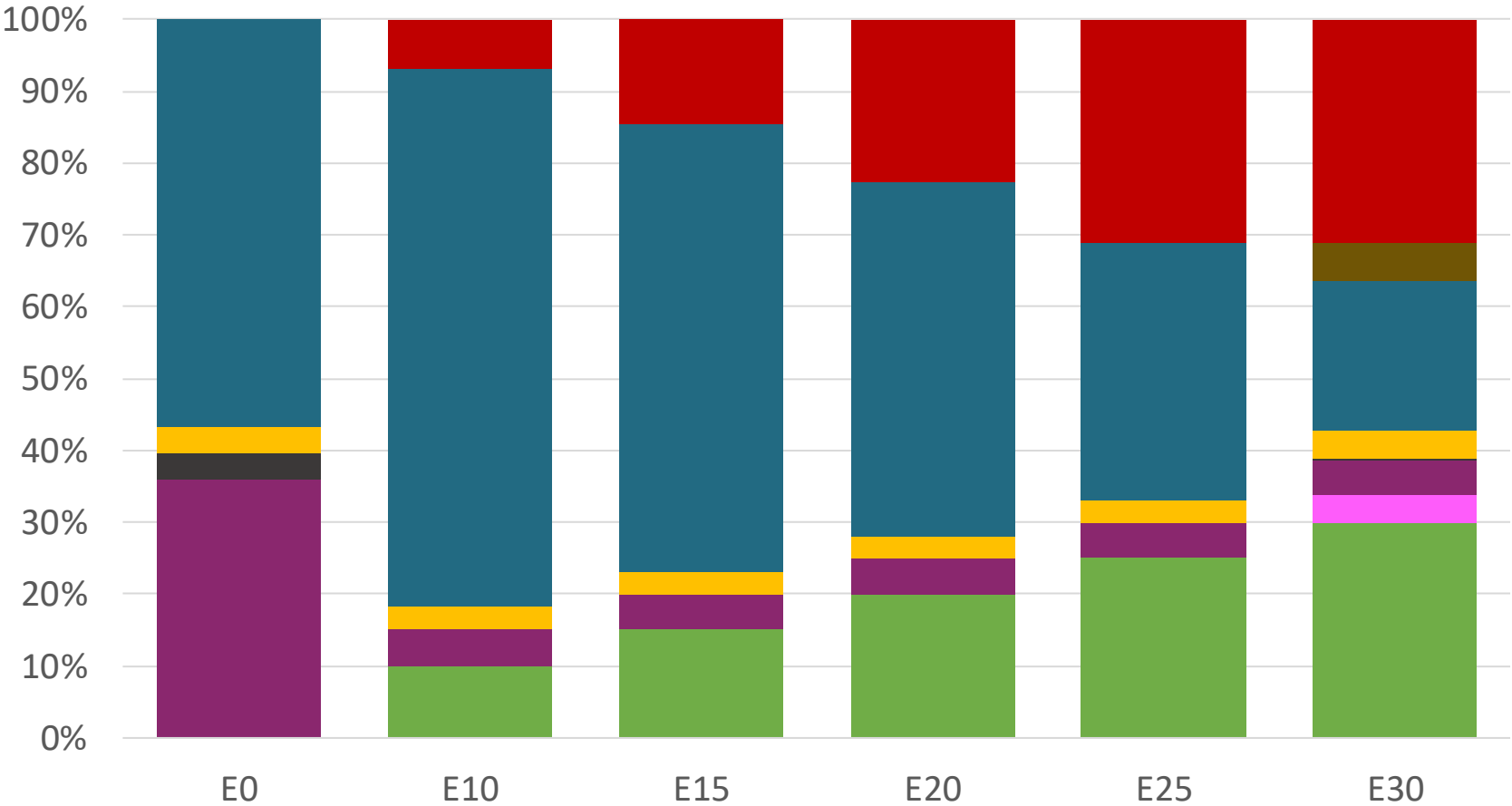
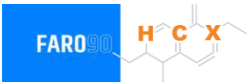
Source: Faro90

Nicaragua – Regular – Constant Octane



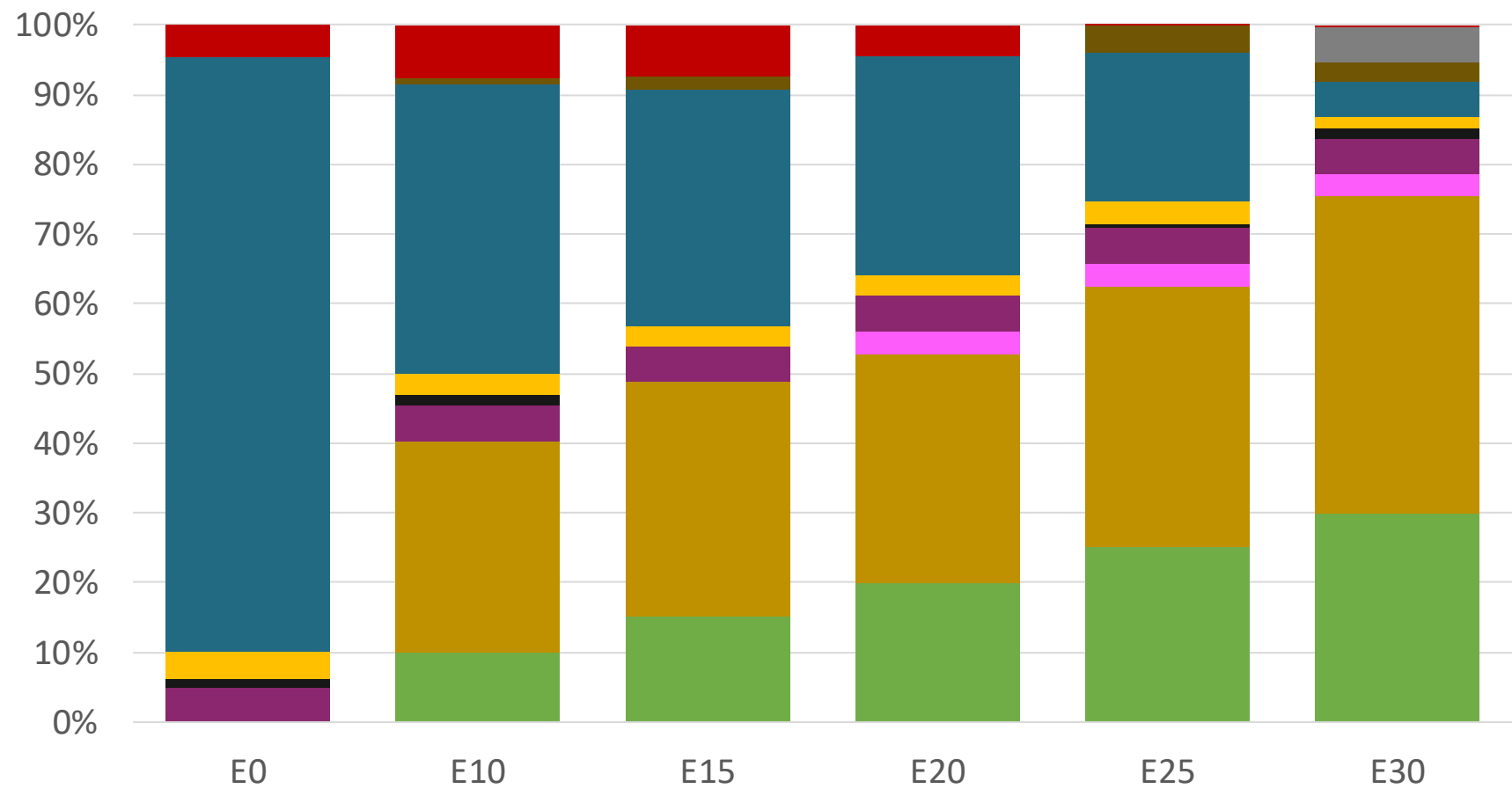
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 1.946	\$ 1.674	\$ 1.523	\$ 1.381	\$ 1.237	\$ 1.266

Nicaragua – Premium – Constant Octane



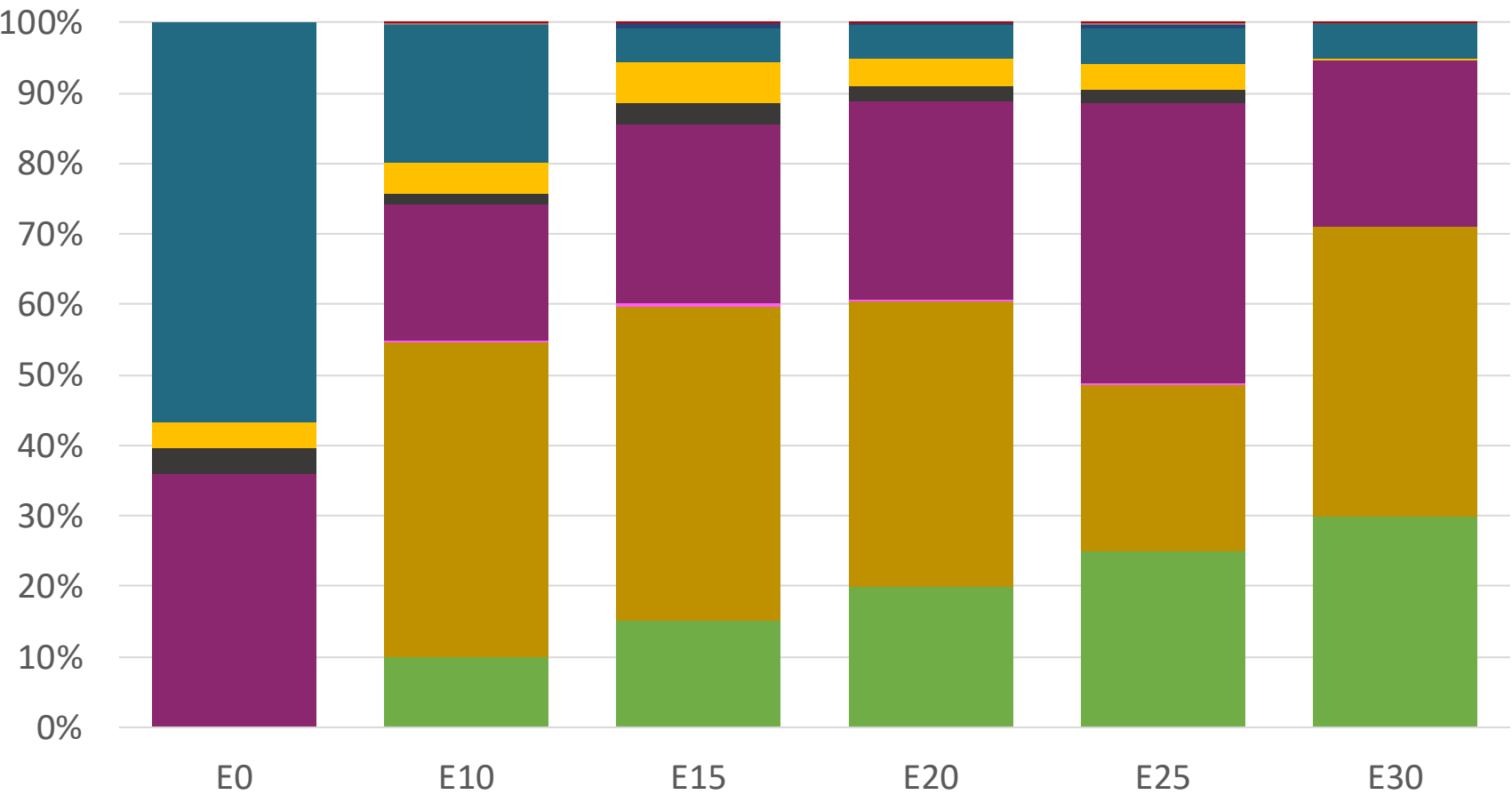
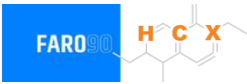
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Nicaragua – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946

Nicaragua – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

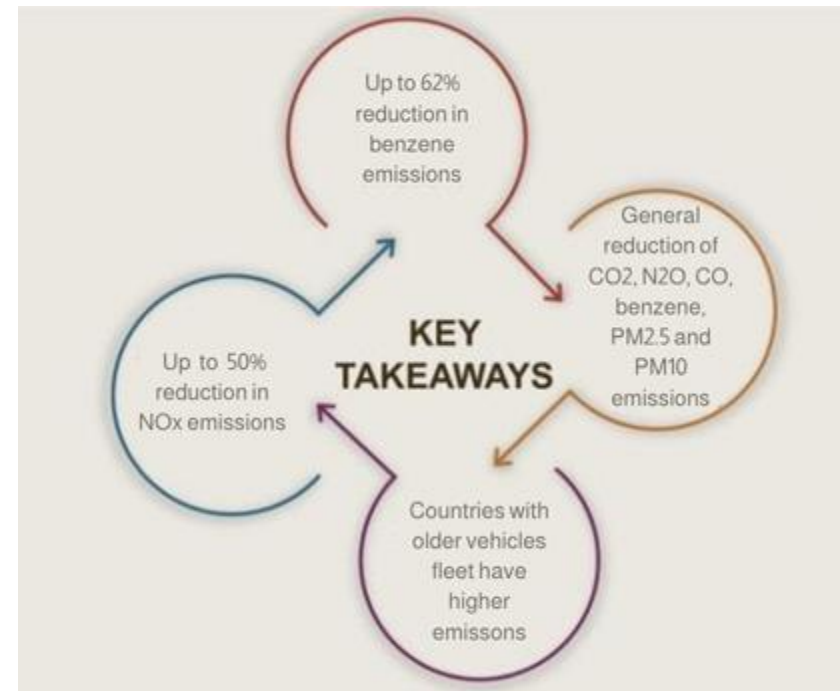
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Vehicular Fleet a gasolina en Nicaragua

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	3,110				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	10,906				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	76,779				
		Euro III		> 8 years, > 160 tkm	50,761				
		Euro IV		> 8 years, > 160 tkm	33,980				
		Euro V / Euro 6		> 8 years, > 160 tkm	20,722				
				4-8 years, 80-160 tkm	11,153				
		< 4 years, < 80 tkm		5,762					
	Electric			Any	111				
	Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	6,337			
Single FI-Pt		3-Ways	> 8 years, > 160 tkm		55,625				
Multi FI-Pt		3-Ways	> 8 years, > 160 tkm		122,643				
		Euro III	> 8 years, > 160 tkm		52,413				
		Euro IV	> 8 years, > 160 tkm		30,359				
		Euro V / Euro 6	> 8 years, > 160 tkm		22,640				
			4-8 years, 80-160 tkm		21,967				
		< 4 years, < 80 tkm	31,268						
Electric			Any		179				
Small Engines		FI 4-cycles	None	PCV	> 8 years, > 160 tkm	10,569			
	Euro II		> 8 years, > 160 tkm		12,856				
	Euro III		> 8 years, > 160 tkm		138,325				
	Euro IV		> 8 years, > 160 tkm		30,294				
	Euro V		4-8 years, 80-160 tkm		103,676				
			4-8 years, 80-160 tkm		42,994				
		< 4 years, < 80 tkm	309,233						
	Electric			Any	556				
	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	3,917				
	FI	3-Ways		> 8 years, > 160 tkm	55,910				
Euro III		> 8 years, > 160 tkm		17,999					
Euro IV		> 8 years, > 160 tkm		8,673					
Euro V / Euro 6		> 8 years, > 160 tkm		5,015					
		4-8 years, 80-160 tkm		19,823					
		< 4 years, < 80 tkm		12,569					
Electric				Any	219				

Vehicular Fleet: **1,204,373**

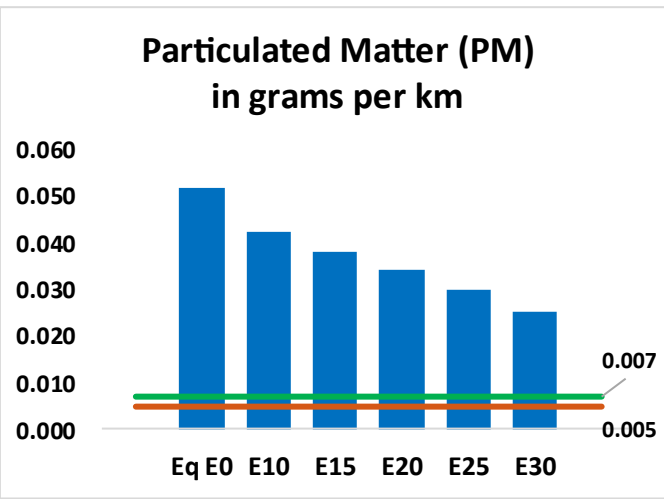
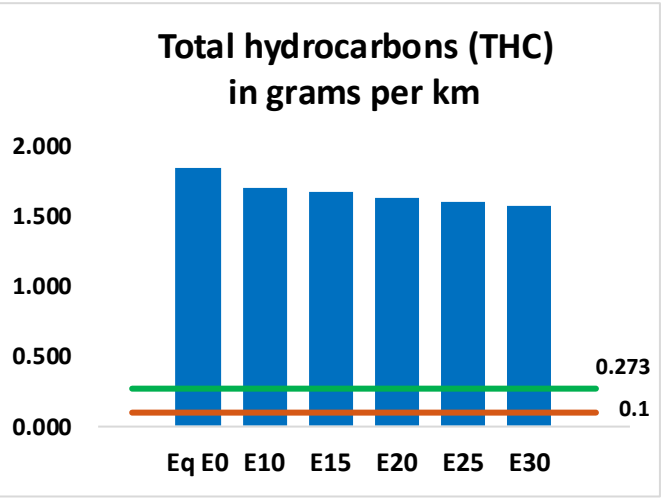
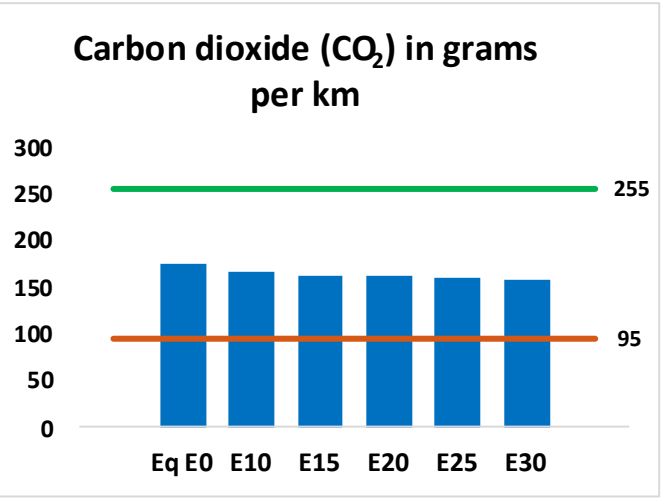
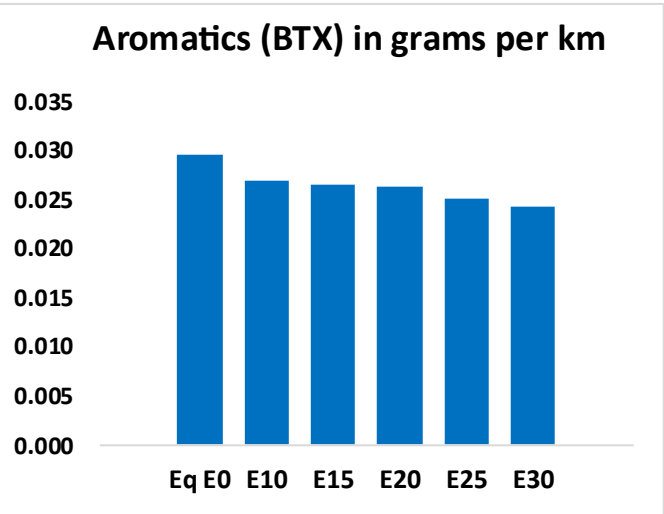
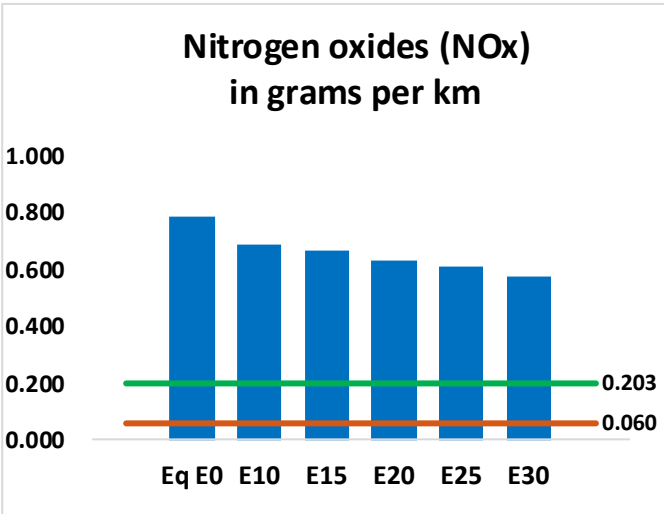
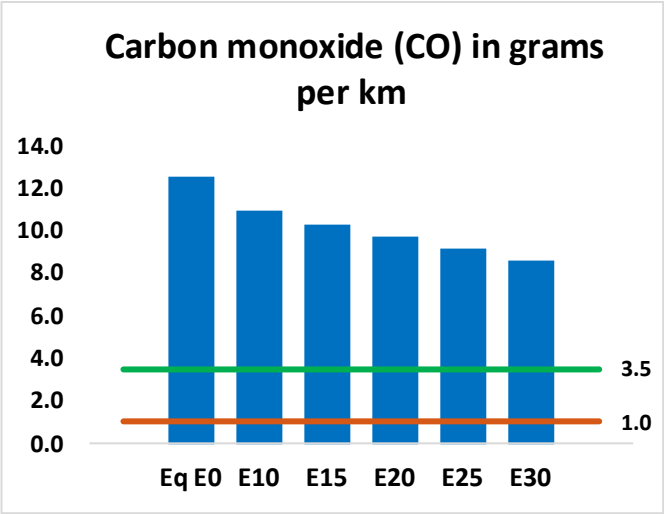
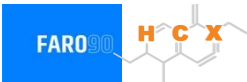
Average Age: **11 years**

Motorcycles: **56%**

Source: ine.gob.hn, analysis Faro 90

Nicaragua - Vehicular Emissions

United States
Euro 6



Nicaragua - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	270,015	252,688	235,362	221,833	208,527	198,388	185,823	-13%	-23%
CO2	3,775,039	3,677,967	3,586,287	3,514,184	3,478,505	3,469,136	3,405,190	-5%	-8%
VOC	39,766	38,306	36,846	35,962	35,215	34,684	33,823	-7%	-11%
NOx	16,992	15,690	14,818	14,385	13,676	13,125	12,474	-13%	-20%
SOx	44	41	37	35	32	29	26	-15%	-28%
NH3	1,513	1,498	1,483	1,490	1,507	1,518	1,528	-2%	0%
N2O	68	67	67	68	69	70	70	-1%	2%
CH4	8,884	8,884	8,884	9,017	9,213	9,382	9,541	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	157	147	137	130	122	116	108	-13%	-22%
Acetaldehyde	395	453	664	1,012	1,379	1,575	1,861	68%	249%
Formaldehyde	1,488	1,446	1,686	1,980	2,074	2,295	2,498	13%	39%
Benzene	639	614	582	573	568	543	526	-9%	-11%
PM2.5	225	204	184	166	148	129	110	-18%	-34%
PM10	893	812	730	659	589	514	436	-18%	-34%

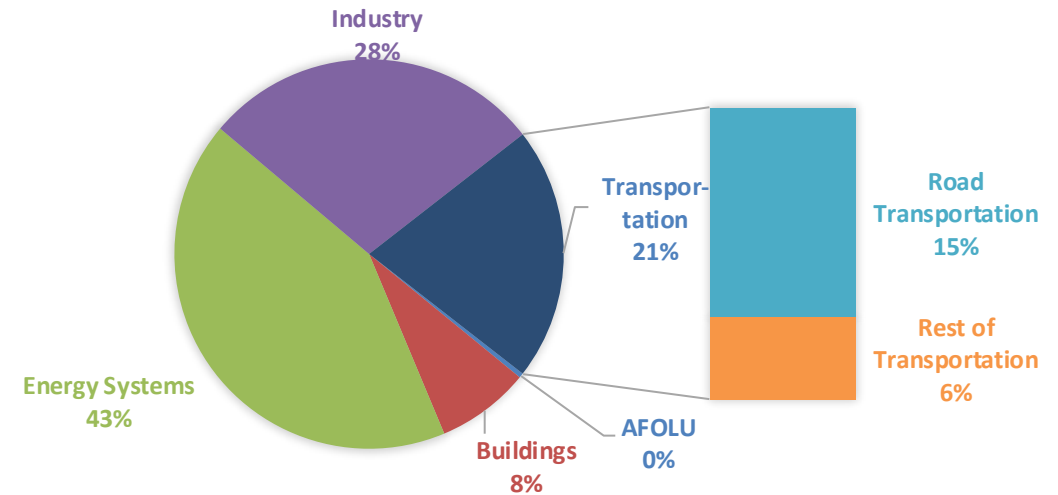
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

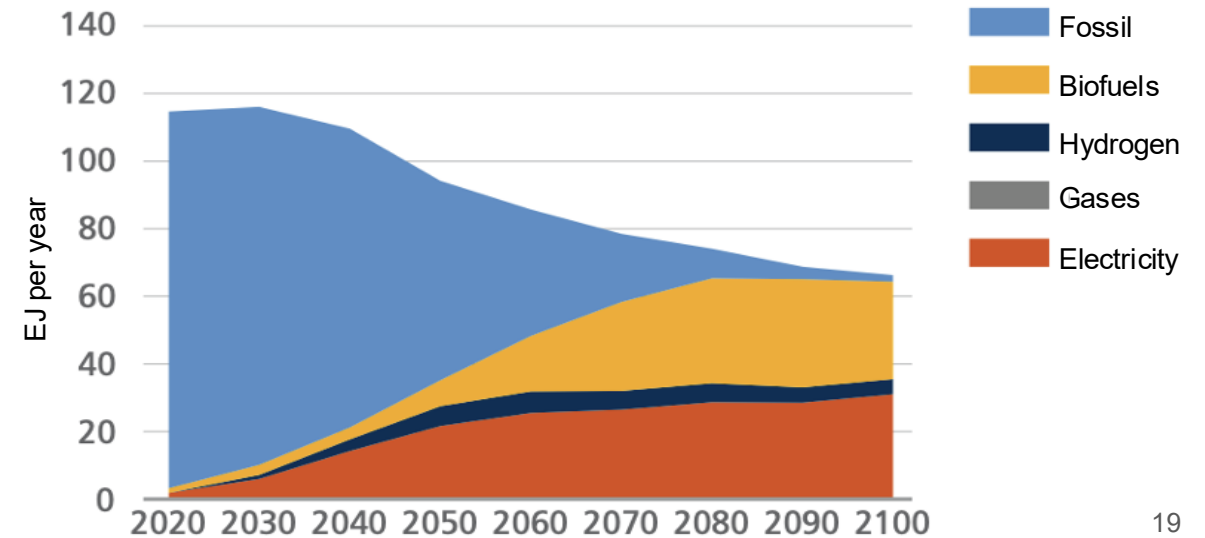
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

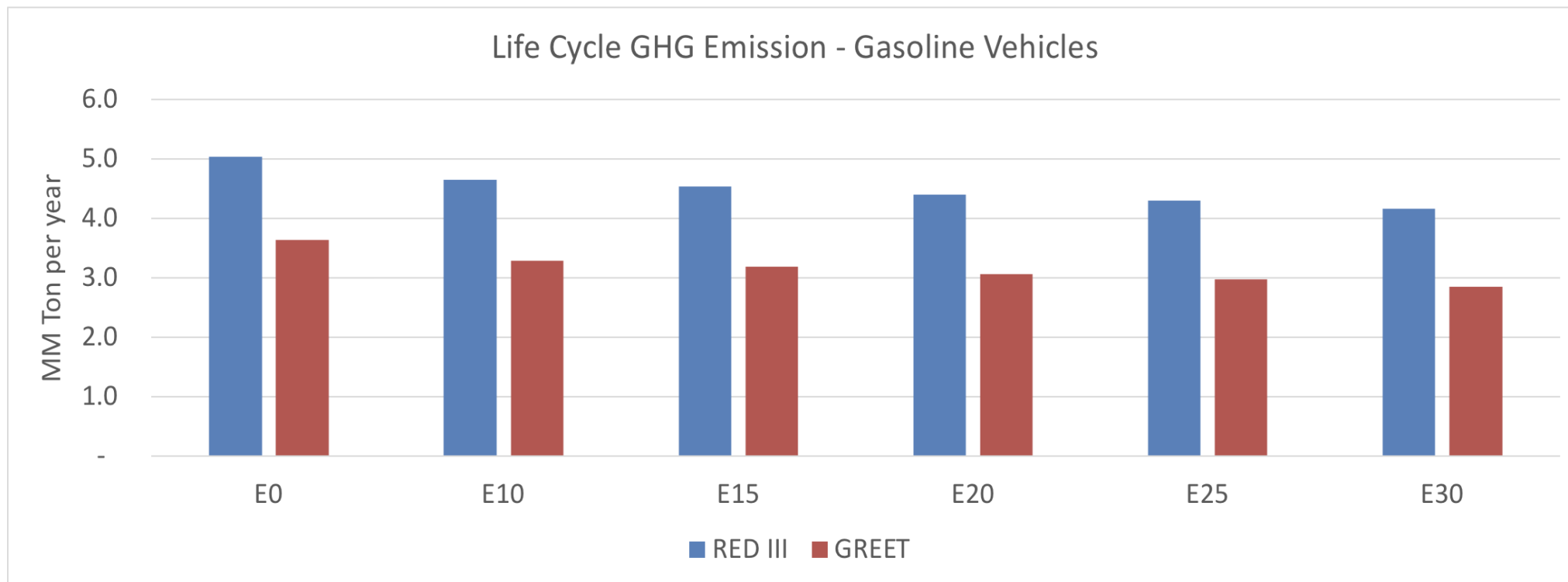
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Nicaragua – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	41.1
2035 Target (MMT/year)	28.8
Delta	12.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.6%	15.9%	18.4%	21.9%
RED II/III	0.0%	7.7%	10.0%	12.5%	14.6%	17.3%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	2.9%	3.7%	4.7%	5.4%	6.5%
RED II/III	0.0%	3.1%	4.1%	5.1%	6.0%	7.0%